

Bahar Viniche

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EDUCATION

Postdoctoral Fellow in Operations Management

Rotman School of Management, University of Toronto
Working with Prof. Sheng Liu and Prof. Nooshin Salari.

Toronto, Canada
September 2025 - Current

PhD in Transportation Engineering and Management

York University and Rotman School of Management, University of Toronto

Toronto, Canada
2021 – 2025

Thesis: Parametric Design of Multi-Echelon Last-Mile Delivery Systems Using Emerging Technologies.

Nominated for the Best Thesis Award in Transportation Engineering and Management (TBA Feb 2026).

Supervisors: Prof. Mehdi Nourinejad, Prof. Opher Baron, Prof. Oded Berman.

Researcher at Interactive-OR lab: Interactive last-mile logistics platforms using data analysis, ML, and Operations Research.

MSc in Air Transportation

Sharif University of Technology

Tehran, Iran
2016 – 2018

Thesis: Airline Development Model based on Sub-Systems Dependencies. Graduated with Distinction. (Thesis Grade: Excellent)

B.S.c in Aerospace Engineering

Sharif University of Technology

Tehran, Iran
2011 – 2015

Graduated with Distinction (Thesis Grade: Excellent)

PUBLICATIONS AND PRESENTATIONS

Journal Publications

Recipient-Dependent Last-Mile Delivery Routing With Autonomous Vehicle Applications.

Bahar D Viniche, Mehdi Nourinejad, Opher Baron, Oded Berman. *Major Revision at Transportation Science Journal*

Parametric Design of Time-Sensitive Routing With Recipient-Dependent Contributions.

Bahar D Viniche, Mehdi Nourinejad, Opher Baron, Oded Berman. *Published at Transportation Research: Part C Journal*

Tactical Routing and Fleet Planning in Drone-Assisted Last-Mile Delivery.

Bahar D Viniche, Mehdi Nourinejad, Opher Baron, Oded Berman. *Under Revision at Operations Research Journal*

Mobile Network Design for Automated External Defibrillators Using Ride-Hailing Platforms.

Bahar D Viniche, Sheng Liu, Nooshin Salari. *Working Paper*

Equitable Territory Planning for Last-Mile Routing with Temporal Flexibility

Bahar D Viniche, Ahana Malhorta, Elkafi Hassini, Mehdi Nourinejad. *Working Paper*

Evaluating Drone-Delivery Efficiency in Different Urban Settings Using GNNs.

Bahar D Viniche, Mehdi Nourinejad, Opher Baron. *Working Paper*

Conference Presentations

Tactical Routing and Fleet Planning in Drone-Assisted Last-Mile Delivery

MSOM Conference, 2025, London, UK

Evaluating Drone-Delivery Efficiency in Different Urban Settings Using Graph Neural Networks (GNNs)

18th INFORMS Computing Society (ICS) Conference, 2025, Toronto, Canada

Strategic Fleet Composition and Efficiency Prediction for Drone-Enabled Last-Mile Delivery.

INFORMS Annual Meeting, 2024, Seattle, US

Tactical Fleet Planning in Drone-Enabled Deliveries and Predicting Drone Delivery Efficiency in Urban Areas using GNNs.

Purdue Operations Conference, 2024, West Lafayette, US

Autonomous Vehicle Applications in Recipient-Dependent Deliveries.

Canadian Operational Research Society (CORS), 2023, Montreal, Canada

Tactical Fleet Planning in Drone-Enabled Deliveries.

INFORMS Annual Meeting, 2023, Phoenix, US

Fleet Composition Optimization in Drone-Enabled Deliveries.

102nd Annual meeting of Transportation Research Board (TRB), 2023, Washington D.C., US

Fleet Composition Optimization in Drone-Enabled Deliveries.

Canadian Operational Research Society (CORS), 2022, Vancouver, Canada

Airline Dynamic Modeling for Uniform Development Based on Sub-Systems Dependencies.

Third Annual Workshop on System Dynamics in Transportation Modelling, 2020, Palermo, Italy

TEACHING EXPERIENCE

Instructor at Rotman School of Management

Scheduled Instructor, Operations Management, Rotman Commerce (two sessions)

Fall 2026

University of Toronto

Teaching Assistant at Rotman School of Management

Statistics for Management, Tools for Probabilistic Models and Prescriptive Analytics, Business Operations Executive MBA (EMBA) Program

2025 – 2026

University of Toronto

Teaching Assistant at Lassonde School of Engineering

Teaching Assistant Excellence Award in Civil Engineering

Capstone Design Project, Project Management, Civil Engineering Design Project, Material and Design, Mysteries of Everyday Materials, Traffic Simulation Modelling.

2021 – 2025

York University

Teaching Assistant at Graduate School of Business and Management

System Dynamics (Graduate Course).

2017 – 2018

Sharif University of Technology

Teaching Assistant at Mechanics and Aerospace Engineering Departments

CAD, Aircraft Performance (Graduate Course), Mechanics of Materials.

2015 – 2018

Sharif University of Technology

Volunteer Tutor at Yarigaran Group

Providing free education to students from underserved families.

2011 – 2013

Sharif University of Technology

INDUSTRY EXPERIENCE

Research and Development Engineer

Sepehran Airline (FlySepehran)

Aug 2019 – Jul 2020

Tehran, Iran

•Conducted research on airline cost management, focusing on optimizing maintenance department expenses. •Using Matlab for cost analysis and statistical modeling. •Contributed to a major project on airline reliability, utilizing data-driven approaches to identify cost-saving opportunities. •Analyzed airline operations through statistical modeling and optimization techniques, developing insights for sustainable growth and long-term operational improvements.

Aviation Intern

Isfahan International Airport

Jun 2014 – Aug 2014

Isfahan, Iran

•Analyzed operational workflows to understand the logistical challenges and decision-making processes in time-sensitive transportation environments. •Gained firsthand experience in the high-speed operations of the control tower.

HONORS AND AWARDS

Best Dissertation Prize Nomination

Lassonde School of Engineering, York University

Toronto, Canada

2025

Teaching Assistant Excellence Award in Civil Engineering

Lassonde School of Engineering, York University

Toronto, Canada

2025

Women in Transportation Graduate Student Award

WTS Women in Transportation Society

Toronto, Canada

2023

Graduate Representative - The Learning Curriculum and Students (LCS) Committee

Lassonde School of Engineering, York University

Toronto, Canada

2023 – 2024

Graduate Representative - Equity, Diversity, and Inclusion (EDI) Subcommittee

York University

Toronto, Canada

2021 – 2022

Academic Excellent Award

York University Faculty of Graduate Studies

Toronto, Canada

2021 - 2025

SELECTED PROJECTS

Equitable Territory Planning for Last-Mile Routing

Toronto, Canada

Project Lead in Academic-Industry Collaboration with Purolator

2024

Developed a strategic territory-planning framework balancing delivery efficiency, driver workload equity, and route familiarity using Purolator operational data. Conducted in collaboration with Smart Freight Centre (SFC) and DeGroote School of Business at McMaster University; results presented at the Purolator Innovation Showcase and SFC Symposium.

Drone Sidekick – Drone-Assisted Routing and Fleet Planning Tool

Toronto, Canada

Interactive Decision-Support Tool

2023

Designed analytical models for optimal truck-and-drone fleet composition and co-developed an interactive web-based platform enabling scenario analysis on real urban networks, bridging theoretical optimization with deployable logistics planning.

Multimodal Transit Improvement

Toronto, Canada

Teaching Assistant and Project Lead

2022

Advised and led a two-semester capstone design project with a 10-student team focused on improving multimodal transit in Southeast Toronto. Guided traffic simulation using VISSIM and ensured technically rigorous, practice-oriented final deliverables.

ET3 – Estimated Travel Time Tool (Vision Zero)

Waterloo, Canada

Project Manager in Academic-Industry Collaboration for Public-Sector Deployment

2022

Led the development and deployment of the ET3 tool for travel-time estimation under varying speed limits, supporting road-safety analysis and public engagement for the City of Waterloo. Coordinated analytics and software teams in a joint project between York University and TraffMobility Engineering Inc.

TECHNICAL SKILLS

Programming Languages and Software:

Python, Gurobi, Matlab, Wolfram Mathematica, JavaScript, HTML, CPLEX, Stata, LATEX, AnyLogic, PowerBI, Vensim, Vissim, ArcGIS.

SERVICE RECORDS

Session Chair

Oct 2023

2023 INFORMS Annual Meeting

Phoenix, US

Session Chair

Jun 2023

2023 Canadian Operation Research Conference (CORS)

Montreal, Canada

Mentor in the Hack ITE York University

Feb 2022

Institute of Transportation Engineers - York University Student Chapter

Toronto, Canada

Moderator in CLUE Symposium

Jan 2022

City Logistics and Urban Economics (CLUE)

Toronto, Canada

CLUE Research-in-Progress Monthly Cafe

Jan 2022 – Jan 2024

City Logistics and Urban Economics (CLUE)

Toronto, Canada

SFC Graduate Research Assistance

Jan 2021 – 2025

Smart Freight Center

Toronto, Canada

Central Council Member of Sharif University Mountaineering Group

Sep 2011 – Aug 2016

Sharif University of Technology

Tehran, Iran

EXTRACURRICULAR INTERESTS

Reading: Avid reader. INFORMS book club and WORMS book club member.

Climbing and Hiking: Certified mountaineer and rock climber (top-rope).

Music: Playing Piano and Harmonica.

REFERENCES

Dr. Mehdi Nourinejad

Assistant Professor, Lassonde School of Engineering, York University

PhD Supervisor

mehdi.nourinejad@lassonde.yorku.ca
+1 416-736-2100 ext. 44080

Dr. Opher Baron

Distinguished Professor, Joseph L. Rotman School of Management, University of Toronto

PhD Co-Supervisor

opher.baron@rotman.utoronto.ca
+1 416-978-4164

TEN SELECTED PUBLICATIONS AND CONTRIBUTIONS

- 1. Tactical Routing and Fleet Planning in Drone-Assisted Last-Mile Delivery.** Viniche, B. D., Baron, O., Berman, O., & Nourinejad, M. *Submitted to Manufacturing & Service Operations Management*. Role: Lead author. Developed the optimization model, algorithms, analytical and numerical analysis, and manuscript. The work studies truck-drone fleet composition and routing under operational constraints such as drone range, payload, and synchronization.
- 2. Recipient-Dependent Last-Mile Delivery Routing With Autonomous Vehicle Applications.** Viniche, B. D., Nourinejad, M., Baron, O., & Berman, O. *Under second-round review at Transportation Science*. Role: Lead author. Responsible for conceptual development, model formulation, coding, numerical analysis, and writing. The study develops analytical foundations for AV-enabled recipient-dependent last-mile delivery systems.
- 3. Parametric Design of Time-Sensitive Routing With Recipient-Dependent Contributions.** Viniche, B. D., Baron, O., Berman, O., & Nourinejad, M. (2025). *Transportation Research Part C: Emerging Technologies*, 178, 105206. Role: Lead author. Developed the research concept, analytical model, computational analysis, and manuscript. This PhD work extends recipient-dependent routing to time-sensitive delivery settings and supports strategic design of emerging last-mile systems.
- 4. Mobile Network Design for Automated External Defibrillators Using Ride-Hailing Platforms.** Viniche, B. D., Liu, S., & Salari, N. *Working paper*. Role: Lead researcher in postdoctoral work. Developing data-driven models for hybrid static-mobile AED networks using emergency incident data, AED location data, and ride-hailing mobility data to improve emergency-response accessibility.
- 5. Drone Sidekick – Interactive Drone Delivery Tool.** *Interactive decision-support platform for truck-and-drone delivery scenario analysis*. Role: Researcher and co-developer. Designed the analytical models and algorithms supporting the tool and contributed to translating theoretical optimization into an interactive platform using real urban networks.
- 6. Equitable Territory Planning for Last-Mile Routing with Temporal Flexibility.** Viniche, B. D., Malhotra, A., Hassini, E., & Nourinejad, M. *Working paper; presented at Purolator Innovation Showcase*. Role: Lead researcher. Developed the conceptual framework and mathematical model for balancing delivery efficiency, driver workload equity, and route familiarity using operational delivery data.
- 7. Evaluating Drone-Delivery Efficiency in Different Urban Settings Using Graph Neural Networks.** Viniche, B. D., Nourinejad, M., & Baron, O. *Working paper; presented at INFORMS Computing Society Conference*. Role: Lead researcher. Developed the research framework combining graph neural networks, urban network data, and drone-delivery performance measures to predict where drone delivery can provide operational benefits.
- 8. ET3 – Estimated Travel Time Tool for Vision Zero.** *Technology-transfer project for the City of Waterloo in collaboration with TraffMobility Engineering Inc.* Role: Project manager. Led development and deployment of a travel-time estimation tool under varying speed limits, coordinated analytics and software teams, and contributed to modeling modules supporting road-safety analysis and public engagement.
- 9. Airline Dynamic Modeling for Uniform Development Based on Sub-Systems Dependencies.** Viniche, B. D., & Malaek, S. M. B. *Third Annual Workshop on System Dynamics in Transportation Modelling, Palermo, Italy, 2020*. Role: Lead author and presenter. Developed system-dynamics modeling from my MSc work to study airline development through interdependent operational subsystems, reflecting my early research foundation in transportation systems and aviation operations.
- 10. EDI Challenges and Opportunities in Supply Chain Management.** Dhaness, J., Viniche, B. D., & Nourinejad, M. *Smart Freight Centre Symposium, Toronto, Canada, 2022*. Role: Supervising researcher. Guided the development of a research presentation on equity, diversity, and inclusion challenges in supply chain management, connecting logistics research with broader questions of workforce inclusion and responsible operations.